

Lecture 13: Implicit Differentiation

GOAL: Differentiate functions of the form $F(x, y) = 0$. Use this to find the derivatives of inverse trigonometric functions.

Recall the following:

1. If y is a fcn of x , then by the Chain Rule

$$\bullet \frac{d}{dx}[y^2] = \frac{d}{dx}[(y(x))^2] = 2y \frac{dy}{dx} \quad \checkmark$$

$$\bullet \frac{d}{dx}[\sin(y)] = \frac{d}{dx}[\sin(y(x))] = \cos(y) \frac{dy}{dx}$$

$$\bullet \frac{d}{dx}[\ln y] = \frac{d}{dx}[\ln(y(x))] = \frac{1}{y} \cdot \frac{dy}{dx}$$

2. Taking the derivative does not change equality

$$y = 2x + 1$$

$$y' = 2$$

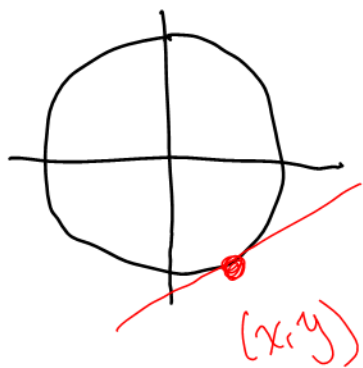
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$$y - 2x - 1 = 0$$

$$y' - 2 + 0 = 0$$

$$y' = 2$$

Q: Find the slope of a line tangent to the circle $x^2 + y^2 = 1$



A: Recognize that y is a function of x .

$$x^2 + y^2 = 1$$

$$x^2 + [y(x)]^2 = 1$$

$$\frac{d}{dx}[x^2 + [y(x)]^2] = \frac{d}{dx}[1]$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Q: Can we rewrite $\frac{dy}{dx}$ solely as a fn of x ?

A: Sometimes!

$$x^2 + y^2 = 1$$

$$y^2 = 1 - x^2$$

$$y = \pm \sqrt{1 - x^2}$$

$$\frac{dy}{dx} = -\frac{x}{y} = -\frac{x}{\pm \sqrt{1-x^2}} = \begin{cases} -\frac{x}{\sqrt{1-x^2}} & \text{if } y > 0 \\ \frac{x}{\sqrt{1-x^2}} & \text{if } y < 0 \end{cases}$$

$x^2 + y^2 = 1$ is an example of an implicit function.
Finding $\frac{dy}{dx}$ is called implicit differentiation.

Explicit Functions

$$y = f(x)$$

$$y = \frac{1}{x}$$

$$y = 2x + 1$$

$$y = \sin(x)$$

Implicit Functions

$$F(x, y) = 0$$

$$xy - 1 = 0$$

$$x^2 + y^2 - 1 = 0$$

$$y \sin(x \cdot y) - \pi = 0$$

Example 1 Find the slope of the line tangent to the elliptic curve

$$y^2 = x^3 - x + 1$$

at $(x, y) = (0, 1)$

$$y^2 = x^3 - x + 1$$

$$\frac{d}{dx}[y^2] = \frac{d}{dx}[x^3 - x + 1]$$

$$2y \frac{dy}{dx} = 3x^2 - 1 + 0$$

$$\frac{dy}{dx} = \frac{3x^2 - 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{(x, y) = (0, 1)} = \frac{3(0)^2 - 1}{2(1)}$$

$$= -\frac{1}{2}$$

$$\begin{aligned} y^2 &= x^3 - x + 1 \\ 1^2 &= 0^3 - 0 + 1 \\ 1 &= 1 \quad \checkmark \end{aligned}$$

Example 3 Find $\frac{dy}{dx}$ for the algebraic curve

$$\frac{d}{dx}(y^3 + yx^2) = \frac{d}{dx}(x^3 + xy^2)$$

$$3y^2 \frac{dy}{dx} + \left[\frac{d}{dx}(x^2) + y(2x) \right] = 3x^2 + \left[y^2 + x(2y) \frac{dy}{dx} \right]$$

$$3y^2 \frac{dy}{dx} + x^2 \frac{dy}{dx} + 2xy = 3x^2 + y^2 + 2xy \frac{dy}{dx}$$

$$3y^2 \frac{dy}{dx} + x^2 \frac{dy}{dx} - 2xy \frac{dy}{dx} = 3x^2 + y^2 - 2xy$$

$$[3y^2 + x^2 - 2xy] \frac{dy}{dx} = 3x^2 + y^2 - 2xy$$

$$\frac{dy}{dx} = \frac{3x^2 + y^2 - 2xy}{3y^2 + x^2 - 2xy}$$

Example 3 Find y' given $xe^y = 9y$

$$xe^y = 9y$$

$$[xe^y]' = [9y]'$$

$$(1) e^y + xe^y y' = 9y'$$

$$xe^y y' - 9y' = -e^y$$

$$[xe^y - 9]y' = -e^y$$

$$y' = -\frac{e^y}{xe^y - 9}$$

$$y' = \frac{e^y}{9 - xe^y}$$

Example 4 Find $\frac{dy}{dx}$ given $10 \sin(2x+3y) = 11xy$

$$10 \sin(2x+3y) = 11xy$$

$$\frac{d}{dx} [10 \sin(2x+3y)] = \frac{d}{dx} [11xy]$$

$$10 \cos(2x+3y) \left[2 + 3 \frac{dy}{dx} \right] = 11y + 11x \frac{dy}{dx}$$

$$20 \cos(2x+3y) + 30 \cos(2x+3y) \frac{dy}{dx} = 11y + 11x \frac{dy}{dx}$$

$$30 \cos(2x+3y) \frac{dy}{dx} - 11x \frac{dy}{dx} = 11y - 20 \cos(2x+3y)$$

$$[30 \cos(2x+3y) - 11x] \frac{dy}{dx} = 11y - 20 \cos(2x+3y)$$

$$\frac{dy}{dx} = \frac{11y - 20 \cos(2x+3y)}{30 \cos(2x+3y) - 11x}$$

Derivatives of $\sin^{-1}$ and $\tan^{-1}$

EX Find $\frac{dy}{dx}$ given $y = \sin^{-1}(x)$

$$y = \sin^{-1}(x)$$

$$\sin(y) = x$$

$$\cos(y) \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos(y)}$$

$$\frac{dy}{dx} = \frac{1}{\cos(\sin^{-1}x)}$$

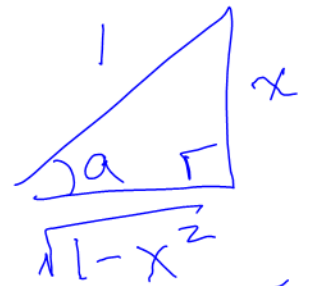
$$\frac{dy}{dx} = \frac{1}{\cos(a)}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

Let

$$a = \sin^{-1}(x)$$

$$\sin(a) = \frac{x}{1}$$



Ex 8 Find $\frac{dy}{dx}$ given $y = \tan^{-1} x$

$$\tan(y) = x$$

$$\sec^2(y) \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2(y)}$$

$$= \frac{1}{[\sec(\tan^{-1}(x))]^2}$$

$$= \frac{1}{1+x^2}$$

$$a = \tan^{-1}(x)$$

$$\tan(a) = \frac{x}{1}$$



Ex Find $\frac{dy}{dx}$ given $\frac{1}{x} + \frac{1}{y} = 1$

Recall $\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = -x^{-2} = -\frac{1}{x^2}$

$$\frac{d}{dx}\left(\frac{1}{x} + \frac{1}{y}\right) = \frac{d}{dx}(1)$$

$$-\frac{1}{x^2} - \frac{1}{y^2} \frac{dy}{dx} = 0$$

$$y^2 \left[\frac{1}{y^2} \frac{dy}{dx} = -\frac{1}{x^2} \right]$$

$$\frac{dy}{dx} = -\frac{y^2}{x^2}$$

Ex Find $\frac{dy}{dx}$ given $\sqrt{x} + \sqrt{y} = 4$

$$\sqrt{x} + \sqrt{y} = 4$$
$$\frac{d}{dx}(\sqrt{x} + \sqrt{y}) = \frac{d}{dx}(4)$$

$$\boxed{\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}}$$

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$$

$$\frac{1}{2\sqrt{y}} \left[\frac{dy}{dx} = -\frac{1}{2\sqrt{x}} \right]$$

$$\frac{dy}{dx} = -\frac{2\sqrt{y}}{2\sqrt{x}} = -\sqrt{\frac{y}{x}}$$